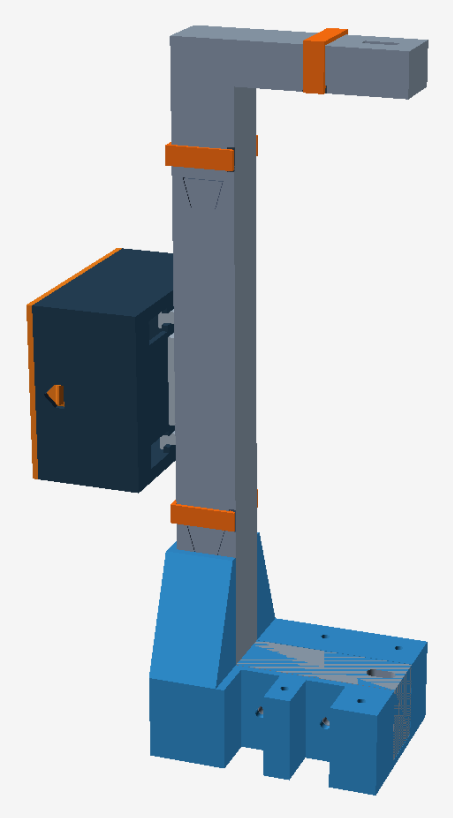
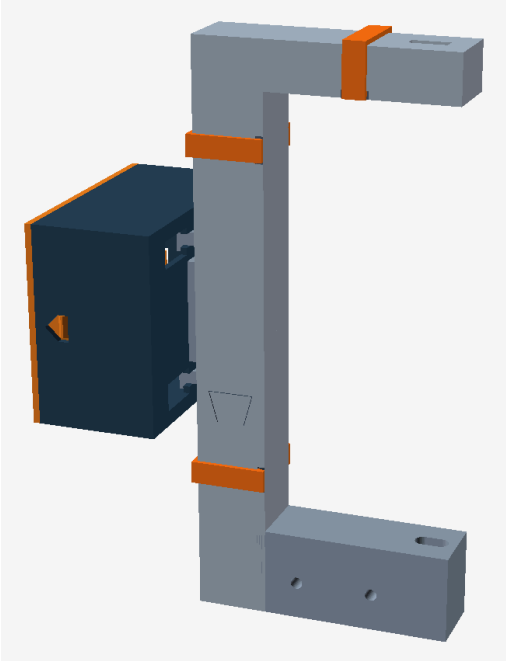
Pinewood U v3 – Assembly Guide

Light barrier holder (Adafruit 2167) for the two-lane pinewood derby track. Start (340 mm, on the inclined ramp) and finish (240 mm, level).



Start unit with foot



Finish unit

1. Parts list

Every unit consists of a back shell (dark, with alignment pins) and a front shell (light, with nut recess). Both shells are always needed. The STL files are in `stl/`.

Part	STL file – Start	Start	STL file – Finish	Finish
Back shell, segments	`start_<hand>_back_shell_seg1..3`	3	`finish_<hand>_back_shell_seg1..2`	2
Front shell, segments	`start_<hand>_front_shell_seg1..3`	3	`finish_<hand>_front_shell_seg1..2`	2
Start foot	`start_<hand>_foot`	1	–	–
Spine clip	`spine_clip`	2	`spine_clip`	2
Upper arm clip	`arm_clip`	1	`arm_clip`	1
Electronics housing	`housing_base`	1	`housing_base`	1
Housing lid	`housing_lid`	1	`housing_lid`	1
Docking lock plug	`dock_lock`	2	`dock_lock`	2

Left/right: `*_right` is the complete mirror image of `*_left`. If the two U-frames of a gate stand on opposite outer sides of the track, each gate needs one left and one right frame, i.e. `start_left`, `start_right`, `finish_left` and `finish_right` in total. If both stand on the same side, use the same hand twice. Clips, housing and lock plugs are not handed.

For the usual track (2 lanes, start and finish, frames facing each other) that is 4 units: 20 shell segments, 2 feet, 8 spine clips, 4 arm clips, 4 housings with lid and 8 lock plugs.

Hardware and tools

- Per unit 2 metal angles that stay on the track permanently (assumed: 20 mm wide, M5 hole, 35 mm hole spacing).
- Finish: 2 × M5×40 with nut. Start: 2 × M5×45 with nut. The nut sits recessed in the front shell.
- Optional, start only: 4 × wood screw 4×30, for support only.
- Glue for PETG (e.g. 2-part epoxy or CA glue with activator), small screwdriver, 8 mm spanner.

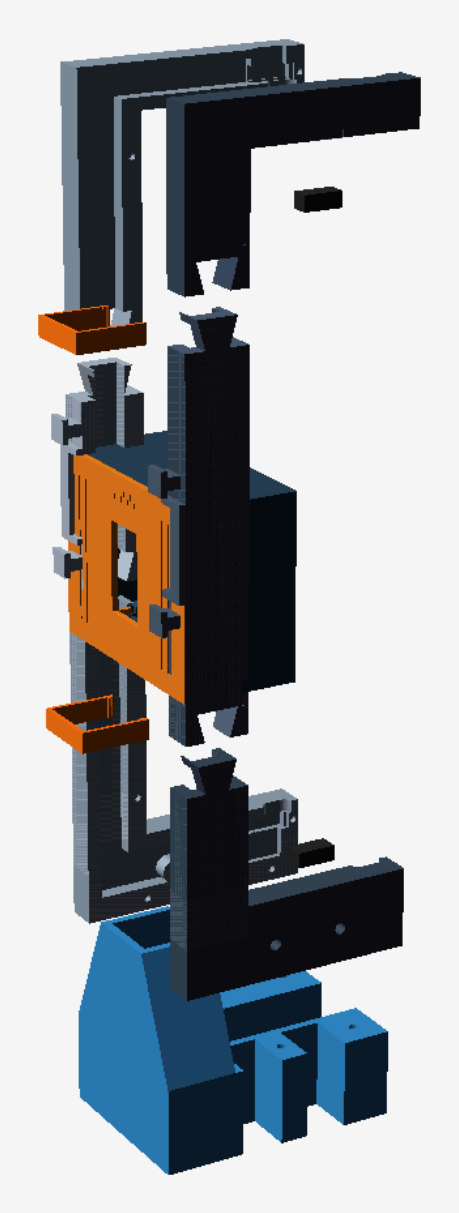
2. Printing

- All STLs are exported in print orientation and need no supports.
- Recommended: PETG, 0.4 mm nozzle, 0.2 mm layers, 4 perimeters, 25–35 % infill.
- Shells: wide outer face on the bed, cable channel and sensor pocket facing up.
- Start foot: bottom on the bed. The horizontal bolt holes are teardrop-shaped (45° roof).
- Largest part: finish front shell segment 1, 127 × 204 mm. Everything fits a 250 mm bed.

3. Test prints first

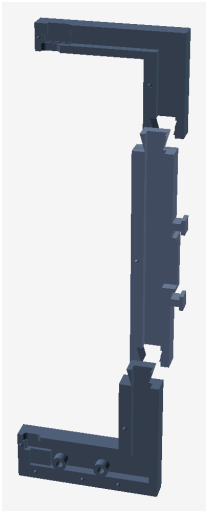
- `test_sensor_seat.stl` (4 small parts): press the sensor into the 1 mm seat of the back shell, add the front shell, check top and bottom. It must sit straight and without wobble. Too loose: `sensor_seat_depth` 1.5–2. Tilted after joining: `sensor_front_relief` 0.25.
- `test_dovetail.stl`: slide the segments together across the joint. If it binds: `fit` 0.35.
- Check the docking on the printed housing. Too tight: `dock_ramp_h` 0.4. Still play: 0.5.

4. Exploded view of the start unit



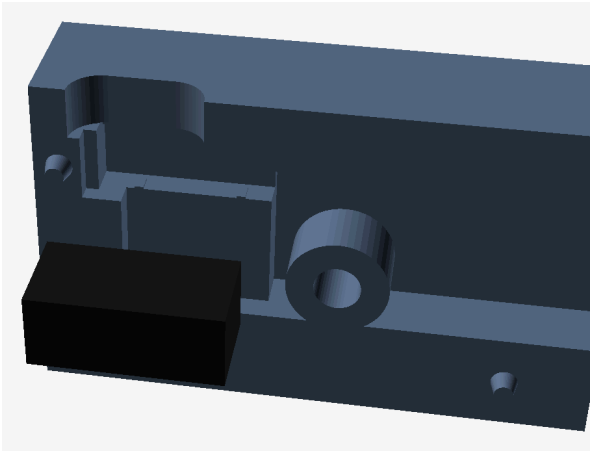
1. Back shell (dark), 3 segments
2. Front shell (light), 3 segments, staggered joints
3. Sensors (black) at the upper and lower arm tips
4. Clips (orange): 2 on the spine, 1 on the upper arm
5. Electronics housing with lid, docked to the back of the spine
6. Start foot (blue), start only

5. Assembly



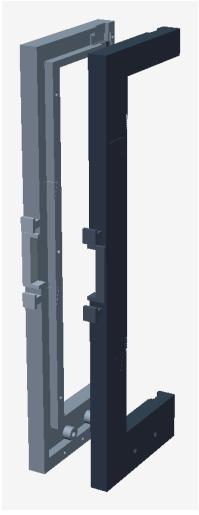
1 Join the back shell

Slide the back shell segments together across the joint (dovetail), dry-fit, then glue. Do the same with the front shell. The joints of the two shells are staggered, so every joint is covered by the other shell.



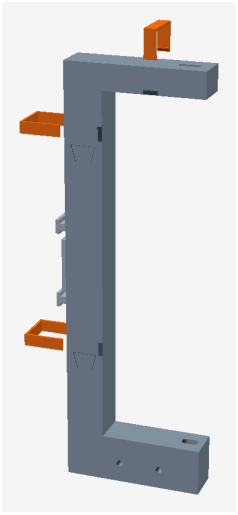
2 Insert the sensors

Press the lens side of each sensor into the 1 mm deep seat of the back shell, top and bottom. Behind it the pocket is open: route the cables through the cable channel to the spine. In the lower arm the cables run **over** the two bolt tubes. To remove a sensor, put a small screwdriver into the notch at the arm tip and lever it out.



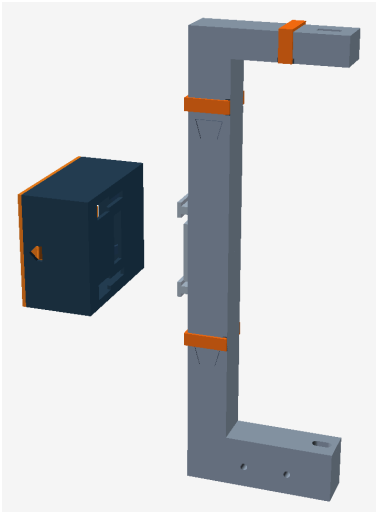
3 Add the front shell

Place the front shell on the alignment pins. Check that both sensors sit straight, then glue over the full surface.



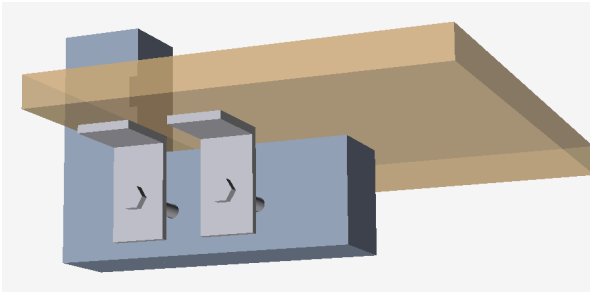
4 Slide on the clips

Slide two clips onto the spine and one onto the upper arm. They hold the shells together while gluing and stay on afterwards.



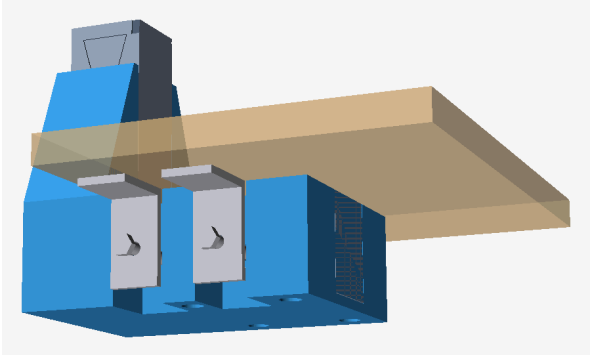
5 Dock the housing

Mount the ESP32 in the housing and route the cables through the passage. Put the housing onto the studs on the spine and slide it 27 mm until it sits firmly on the wedge ramps. Then insert the two lock plugs.



6 Mount the finish

The two metal angles stay on the track permanently. Hold the U against the angles and push 2 × M5×40 from the angle side through angle and U; the nut sits recessed in the front shell and cannot turn.



7 Mount the start

Stand the U in the foot. Drop the nut through the large counterbore in the foot into the recess of the front shell. Put the angles into the flange slots and push 2 × M5×45 through angle, foot and U. The wide flange supports the U sideways on the ramp. Only if it does not hold well enough: add 4 wood screws 4×30 from below through the flange.

6. Swapping and maintenance

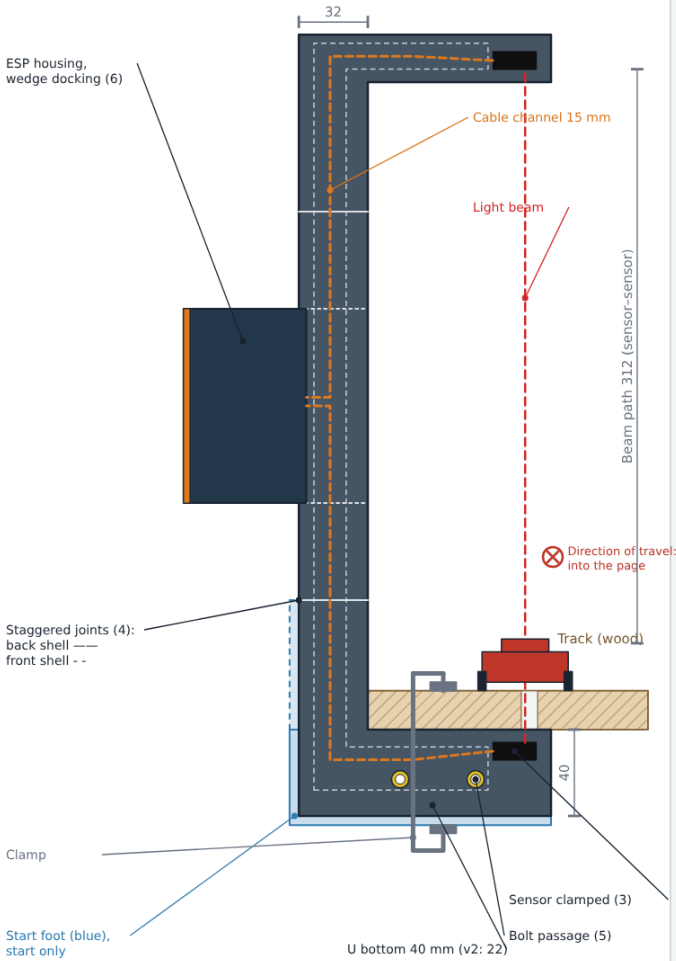
- Swapping: undo 2 bolts and take the U off, at the start together with the foot. Nothing on the track is changed.
- If the start foot is also screwed on, the wood screws must come out too: the foot wraps around the U bottom and the track lies above it.
- All dimensions are in `pinewood_v3_shared.scad`. Run `./export_stls_v3.sh` again after every change.

7. Design sketch

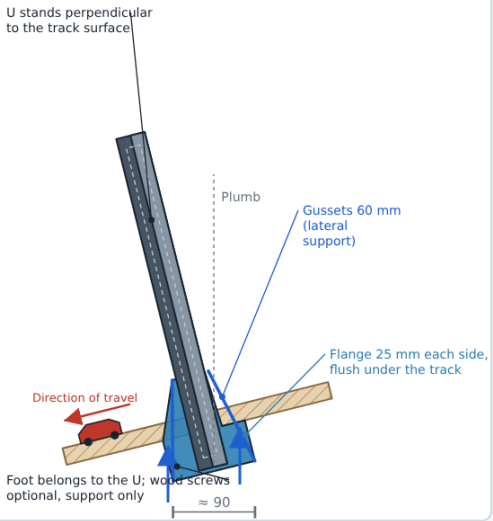
Pinewood U v3 – design sketch (schematic, start variant left)

Dimensions in mm from pinewood_v3_shared.scad. Red dashed = light beam, orange dashed = cable route. Not a scaled manufacturing drawing.

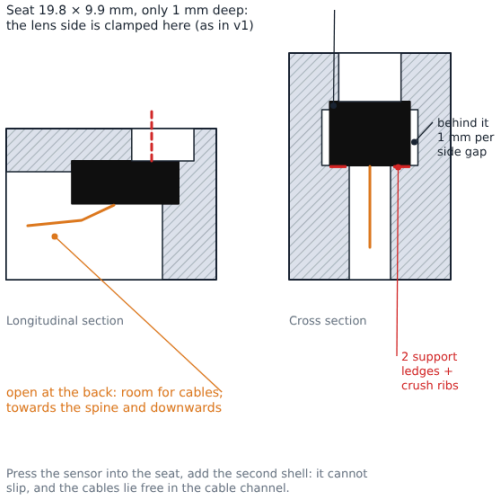
1 Front view: start U at the track



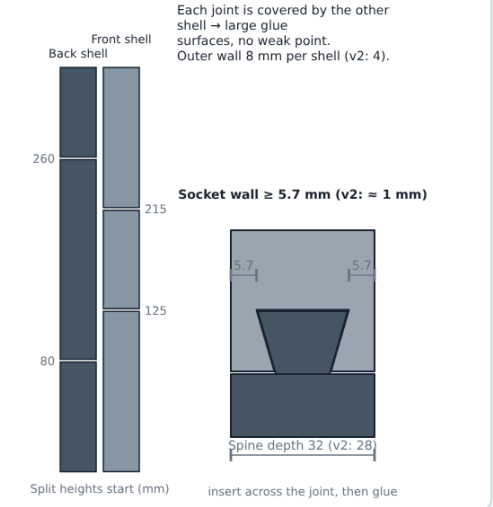
2 Side view start: inclined track



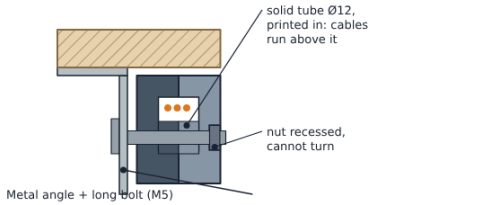
3 Sensor mount (sections of arm tip)



4 Staggered joints + strong dovetail



5 Bolt passage (section lower arm)



6 Housing docking with wedge ramp

